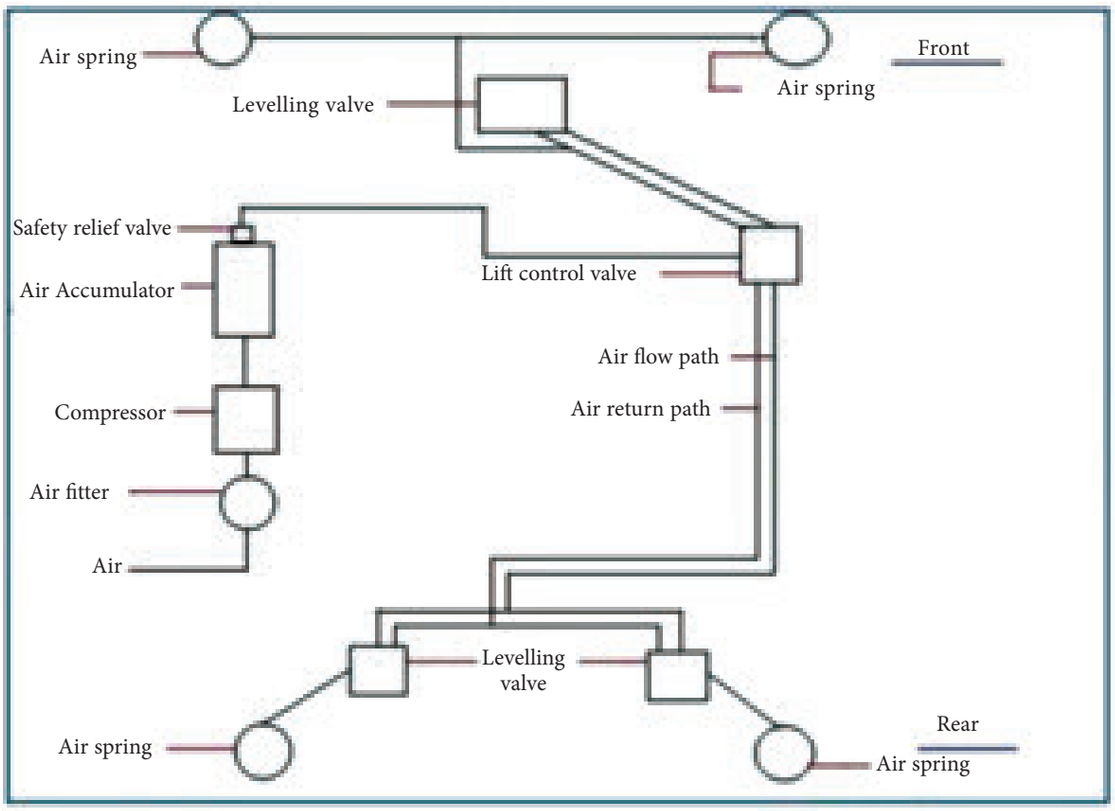


Figure 7.3.3 (2) Air suspension system

7.4 Independent suspension system advantages and disadvantages

Advantages

1. When one side of the wheel is subjected to road disturbance, then this disturbance does not transmit to the other side of the wheel.
2. During cornering, the rollover, the skid, pull on one side etc will not occur.
3. It provides good road holding property.
4. It promotes easy turning of the vehicle and have better control.
5. It ensures good ride quality.
6. It prevents the wheel from turning towards the kingpin axis.
7. In front wheel steering vehicles, it provides space to locate the engine.
8. It reduces the unsprung weight.
9. It controls the effect of steering geometry hence steering is easier.

Disadvantages

1. It has more probable for tyre wear.
2. Comparatively, the cost is high.
3. Requires more maintenance and it is complex.
4. Maintenance cost is high.
5. It requires good quality shock absorber to enhance the ride comfort.
6. Construction is complex.
7. More benefits when it is mounted in the front axle than rear axle.

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